mechanics with ease and clarity. He introduced the concepts of kets, bras, and bra-kets, which will be explained below.

Kets: elements of a vector space
Dirac denoted the state vector ψ by the symbol $|\ \psi\rangle$, which he called a *ket* vector, or simply a ket. Kets belong to the Hilbert (vector) space $\mathcal{H}$, or, in short, to the ket-space.

Bras: elements of a dual space
As mentioned above, we know from linear algebra that a dual space can be associated with every vector space. Dirac denoted the elements of a dual space by the symbol $\langle\ |$, which he called a bra vector, or simply a bra; for instance, the element $\langle\psi\ |$ represents a bra. *Note: For every ket* $|\ \psi\rangle$ *there exists a unique bra* $\langle\psi\ |$ *and vice versa*. Again, while kets belong to the Hilbert space $\mathcal{H}$, the corresponding bras belong to its dual (Hilbert) space $\mathcal{H}_d$.

Bra-ket: Dirac notation for the scalar product
Dirac denoted the scalar (inner) product by the symbol $\langle\ |\ \rangle$, which he called a a *bra-ket*. For instance, the scalar product (ϕ, ψ) is denoted by the bra-ket $\langle\phi\ |\ \psi\rangle$:

$$(\phi, \psi) \quad \longrightarrow \quad \langle\phi\ |\ \psi\rangle. \tag{2.24}$$

Note: When a ket (or bra) is multiplied by a complex number, we also get a ket (or bra).

Remark: In wave mechanics we deal with wave functions $\psi(\vec{r}, t)$, but in the more general formalism of quantum mechanics we deal with abstract kets $|\ \psi\rangle$. Wave functions, like kets, are elements of a Hilbert space. We should note that, like a wave function, a ket represents the system completely, and hence knowing $|\ \psi\rangle$ means knowing all its amplitudes in all possible representations. As mentioned above, kets are independent of any particular representation. There is no reason to single out a particular representation basis such as the representation in the position space. Of course, if we want to know the probability of finding the particle at some position in space, we need to work out the formalism within the coordinate representation. The state vector of this particle at time t will be given by the spatial wave function $\langle\vec{r}, t\ |\ \psi\rangle = \psi(\vec{r}, t)$. In the coordinate representation, the scalar product $\langle\phi\ |\ \psi\rangle$ is given by

$$\langle\phi\ |\ \psi\rangle = \int \phi^*(\vec{r}, t)\,\psi(\vec{r}, t)\,d^3r. \tag{2.25}$$

Similarly, if we are considering the three-dimensional momentum of a particle, the ket $|\ \psi\rangle$ will have to be expressed in momentum space. In this case the state of the particle will be described by a wave function $\psi(\vec{p}, t)$, where $\vec{p}$ is the momentum of the particle.

Properties of kets, bras, and bra-kets

- **Every ket has a corresponding bra**

  To every *ket* $|\ \psi\rangle$, there corresponds a unique *bra* $\langle\psi\ |$ and vice versa:

  $$|\ \psi\rangle \quad \longleftrightarrow \quad \langle\psi\ |\ . \tag{2.26}$$

  There is a one-to-one correspondence between bras and kets:

  $$a\ |\ \psi\rangle + b\ |\ \phi\rangle \quad \longleftrightarrow \quad a^*\langle\psi\ | + b^*\langle\phi\ |, \tag{2.27}$$

  where a and b are complex numbers. The following is a common notation:

  $$|\ a\psi\rangle = a\ |\ \psi\rangle, \qquad \langle a\psi\ | = a^*\langle\psi\ |\ . \tag{2.28}$$

- **Properties of the scalar product**

  In quantum mechanics, since the scalar product is a complex number, the ordering matters a lot. We must be careful to distinguish a scalar product from its complex conjugate; $\langle \psi \mid \phi \rangle$ is not the same thing as $\langle \phi \mid \psi \rangle$:

$$\langle \phi \mid \psi \rangle^* = \langle \psi \mid \phi \rangle. \tag{2.29}$$

  This property becomes clearer if we apply it to (2.21):

$$\langle \phi \mid \psi \rangle^* = \left(\int \phi^*(\vec{r}, t) \psi(\vec{r}, t)\, d^3r \right)^* = \int \psi^*(\vec{r}, t) \phi(\vec{r}, t)\, d^3r = \langle \psi \mid \phi \rangle. \tag{2.30}$$

  When $\mid \psi \rangle$ and $\mid \phi \rangle$ are real, we would have $\langle \psi \mid \phi \rangle = \langle \phi \mid \psi \rangle$. Let us list some additional properties of the scalar product:

$$\langle \psi \mid a_1 \psi_1 + a_2 \psi_2 \rangle = a_1 \langle \psi \mid \psi_1 \rangle + a_2 \langle \psi \mid \psi_2 \rangle, \tag{2.31}$$

$$\langle a_1 \phi_1 + a_2 \phi_2 \mid \psi \rangle = a_1^* \langle \phi_1 \mid \psi \rangle + a_2^* \langle \phi_2 \mid \psi \rangle, \tag{2.32}$$

$$\begin{aligned} \langle a_1 \phi_1 + a_2 \phi_2 \mid b_1 \psi_1 + b_2 \psi_2 \rangle = {} & a_1^* b_1 \langle \phi_1 \mid \psi_1 \rangle + a_1^* b_2 \langle \phi_1 \mid \psi_2 \rangle \\ & + a_2^* b_1 \langle \phi_2 \mid \psi_1 \rangle + a_2^* b_2 \langle \phi_2 \mid \psi_2 \rangle. \end{aligned} \tag{2.33}$$

- **The norm is real and positive**

  For any state vector $\mid \psi \rangle$ of the Hilbert space $\mathcal{H}$, the norm $\langle \psi \mid \psi \rangle$ is real and positive; $\langle \psi \mid \psi \rangle$ is equal to zero only for the case where $\mid \psi \rangle = O$, where O is the zero vector. If the state $\mid \psi \rangle$ is normalized then $\langle \psi \mid \psi \rangle = 1$.

- **Schwarz inequality**

  For any two states $\mid \psi \rangle$ and $\mid \phi \rangle$ of the Hilbert space, we can show that

$$|\langle \psi \mid \phi \rangle|^2 \leq \langle \psi \mid \psi \rangle \langle \phi \mid \phi \rangle. \tag{2.34}$$

  If $\mid \psi \rangle$ and $\mid \phi \rangle$ are linearly dependent (i.e., proportional: $\mid \psi \rangle = \alpha \mid \phi \rangle$, where α is a scalar), this relation becomes an equality. The Schwarz inequality (2.34) is analogous to the following relation of the real Euclidean space

$$|\vec{A} \cdot \vec{B}|^2 \leq |\vec{A}|^2 \, |\vec{B}|^2 . \tag{2.35}$$

- **Triangle inequality**

$$\sqrt{\langle \psi + \phi \mid \psi + \phi \rangle} \leq \sqrt{\langle \psi \mid \psi \rangle} + \sqrt{\langle \phi \mid \phi \rangle}. \tag{2.36}$$

  If $\mid \psi \rangle$ and $\mid \phi \rangle$ are linearly dependent, $\mid \psi \rangle = \alpha \mid \phi \rangle$, and if the proportionality scalar α is real and positive, the triangle inequality becomes an equality. The counterpart of this inequality in Euclidean space is given by $|\vec{A} + \vec{B}| \leq |\vec{A}| + |\vec{B}|$.

- **Orthogonal states**

  Two *kets*, $\mid \psi \rangle$ and $\mid \phi \rangle$, are said to be orthogonal if they have a vanishing scalar product:

$$\langle \psi \mid \phi \rangle = 0. \tag{2.37}$$

- **Orthonormal states**

  Two *kets*, $|\ \psi\rangle$ and $|\ \phi\rangle$, are said to be orthonormal if they are orthogonal and if each one of them has a unit norm:

$$\langle\psi\ |\ \phi\rangle = 0, \qquad \langle\psi\ |\ \psi\rangle = 1, \qquad \langle\phi\ |\ \phi\rangle = 1. \tag{2.38}$$

- **Forbidden quantities**

  If $|\ \psi\rangle$ and $|\ \phi\rangle$ belong to the same vector (Hilbert) space, products of the type $|\ \psi\rangle\ |\ \phi\rangle$ and $\langle\psi\ |\ \langle\phi\ |$ are forbidden. They are nonsensical, since $|\ \psi\rangle\ |\ \phi\rangle$ and $\langle\psi\ |\ \langle\phi\ |$ are neither kets nor bras (an explicit illustration of this will be carried out in the example below and later on when we discuss the representation in a discrete basis). If $|\ \psi\rangle$ and $|\ \phi\rangle$ belong, however, to different vector spaces (e.g., $|\ \psi\rangle$ belongs to a spin space and $|\ \phi\rangle$ to an orbital angular momentum space), then the product $|\ \psi\rangle\ |\ \phi\rangle$, written as $|\ \psi\rangle \otimes |\ \phi\rangle$, represents a tensor product of $|\ \psi\rangle$ and $|\ \phi\rangle$. Only in these typical cases are such products meaningful.

Example 2.3

(**Note:** We will see later in this chapter that kets are represented by column matrices and bras by row matrices; this example is offered earlier than it should because we need to show some concrete illustrations of the formalism.) Consider the following two kets:

$$|\ \psi\rangle = \begin{pmatrix} -3i \\ 2+i \\ 4 \end{pmatrix}, \qquad |\ \phi\rangle = \begin{pmatrix} 2 \\ -i \\ 2-3i \end{pmatrix}.$$

(a) Find the bra $\langle\phi\ |$.
(b) Evaluate the scalar product $\langle\phi\ |\ \psi\rangle$.
(c) Examine why the products $|\ \psi\rangle\ |\ \phi\rangle$ and $\langle\phi\ |\ \langle\psi\ |$ do not make sense.

Solution

(a) As will be explained later when we introduce the Hermitian adjoint of kets and bras, we want to mention that the bra $\langle\phi\ |$ can be obtained by simply taking the complex conjugate of the transpose of the ket $|\ \phi\rangle$:

$$\langle\phi\ | = (2 \quad i \quad 2+3i)\,. \tag{2.39}$$

(b) The scalar product $\langle\phi\ |\ \psi\rangle$ can be calculated as follows:

$$\begin{aligned} \langle\phi\ |\ \psi\rangle &= (2 \quad i \quad 2+3i) \begin{pmatrix} -3i \\ 2+i \\ 4 \end{pmatrix} \\ &= 2(-3i) + i(2+i) + 4(2+3i) \\ &= 7 + 8i. \end{aligned} \tag{2.40}$$

(c) First, the product $|\ \psi\rangle\ |\ \phi\rangle$ cannot be performed because, from linear algebra, the product of two column matrices cannot be performed. Similarly, since two row matrices cannot be multiplied, the product $\langle\phi\ |\ \langle\psi\ |$ is meaningless.

Physical meaning of the scalar product

The scalar product can be interpreted in two ways. First, by analogy with the scalar product of ordinary vectors in the Euclidean space, where $\vec{A} \cdot \vec{B}$ represents the projection of $\vec{B}$ on $\vec{A}$, the product $\langle \phi \mid \psi \rangle$ also represents the projection of $\mid \psi \rangle$ onto $\mid \phi \rangle$. Second, in the case of normalized states and according to Born's probabilistic interpretation, the quantity $\langle \phi \mid \psi \rangle$ represents the probability amplitude that the system's state $\mid \psi \rangle$ will, after a measurement is performed on the system, be found to be in another state $\mid \phi \rangle$.

Example 2.4 (Bra-ket algebra)

Consider the states $\mid \psi \rangle = 3i \mid \phi_1 \rangle - 7i \mid \phi_2 \rangle$ and $\mid \chi \rangle = - \mid \phi_1 \rangle + 2i \mid \phi_2 \rangle$, where $\mid \phi_1 \rangle$ and $\mid \phi_2 \rangle$ are orthonormal.

(a) Calculate $\mid \psi + \chi \rangle$ and $\langle \psi + \chi \mid$.

(b) Calculate the scalar products $\langle \psi \mid \chi \rangle$ and $\langle \chi \mid \psi \rangle$. Are they equal?

(c) Show that the states $\mid \psi \rangle$ and $\mid \chi \rangle$ satisfy the Schwarz inequality.

(d) Show that the states $\mid \psi \rangle$ and $\mid \chi \rangle$ satisfy the triangle inequality.

Solution

(a) The calculation of $\mid \psi + \chi \rangle$ is straightforward:

$$\begin{aligned} \mid \psi + \chi \rangle &= \mid \psi \rangle + \mid \chi \rangle = (3i \mid \phi_1 \rangle - 7i \mid \phi_2 \rangle) + (- \mid \phi_1 \rangle + 2i \mid \phi_2 \rangle) \\ &= (-1 + 3i) \mid \phi_1 \rangle - 5i \mid \phi_2 \rangle. \end{aligned} \tag{2.41}$$

This leads at once to the expression of $\langle \psi + \chi \mid$:

$$\langle \psi + \chi \mid = (-1 + 3i)^* \langle \phi_1 \mid + (-5i)^* \langle \phi_2 \mid = (-1 - 3i) \langle \phi_1 \mid + 5i \langle \phi_2 \mid . \tag{2.42}$$

(b) Since $\langle \phi_1 \mid \phi_1 \rangle = \langle \phi_2 \mid \phi_2 \rangle = 1$, $\langle \phi_1 \mid \phi_2 \rangle = \langle \phi_2 \mid \phi_1 \rangle = 0$, and since the bras corresponding to the kets $\mid \psi \rangle = 3i \mid \phi_1 \rangle - 7i \mid \phi_2 \rangle$ and $\mid \chi \rangle = - \mid \phi_1 \rangle + 2i \mid \phi_2 \rangle$ are given by $\langle \psi \mid = -3i \langle \phi_1 \mid + 7i \langle \phi_2 \mid$ and $\langle \chi \mid = - \langle \phi_1 \mid - 2i \langle \phi_2 \mid$, the scalar products are

$$\begin{aligned} \langle \psi \mid \chi \rangle &= (-3i \langle \phi_1 \mid + 7i \langle \phi_2 \mid) (- \mid \phi_1 \rangle + 2i \mid \phi_2 \rangle) \\ &= (-3i)(-1) \langle \phi_1 \mid \phi_1 \rangle + (7i)(2i) \langle \phi_2 \mid \phi_2 \rangle \\ &= -14 + 3i, \end{aligned} \tag{2.43}$$

$$\begin{aligned} \langle \chi \mid \psi \rangle &= (- \langle \phi_1 \mid - 2i \langle \phi_2 \mid) (3i \mid \phi_1 \rangle - 7i \mid \phi_2 \rangle) \\ &= (-1)(3i) \langle \phi_1 \mid \phi_1 \rangle + (-2i)(-7i) \langle \phi_2 \mid \phi_2 \rangle \\ &= -14 - 3i. \end{aligned} \tag{2.44}$$

We see that $\langle \psi \mid \chi \rangle$ is equal to the complex conjugate of $\langle \chi \mid \psi \rangle$.

(c) Let us first calculate $\langle \psi \mid \psi \rangle$ and $\langle \chi \mid \chi \rangle$:

$$\langle \psi \mid \psi \rangle = (-3i \langle \phi_1 \mid + 7i \langle \phi_2 \mid) (3i \mid \phi_1 \rangle - 7i \mid \phi_2 \rangle) = (-3i)(3i) + (7i)(-7i) = 58, \tag{2.45}$$

$$\langle \chi \mid \chi \rangle = (- \langle \phi_1 \mid - 2i \langle \phi_2 \mid) (- \mid \phi_1 \rangle + 2i \mid \phi_2 \rangle) = (-1)(-1) + (-2i)(2i) = 5. \tag{2.46}$$

Since $\langle \psi \mid \chi \rangle = -14 + 3i$ we have $\mid \langle \psi \mid \chi \rangle \mid^2 = 14^2 + 3^2 = 205$. Combining the values of $\mid \langle \psi \mid \chi \rangle \mid^2$, $\langle \psi \mid \psi \rangle$, and $\langle \chi \mid \chi \rangle$, we see that the Schwarz inequality (2.34) is satisfied:

$$205 < (58)(5) \Longrightarrow \mid \langle \psi \mid \chi \rangle \mid^2 < \langle \psi \mid \psi \rangle \langle \chi \mid \chi \rangle. \tag{2.47}$$